

Experiment 1

Simulating and Comparing Network Topologies

Lab 1 Objectives:

- Understand and simulate different network topologies: Bus, Star, Mesh, and Ring.

Prerequisites: The prerequisites of this lab are-

- Basic understanding of computer networks and topology structures.
- Familiarity with network simulation tools such as Cisco Packet Tracer or NS-3.
- Basic knowledge of network communication principles.

Outcomes: By the end of this lab, students will be able to:

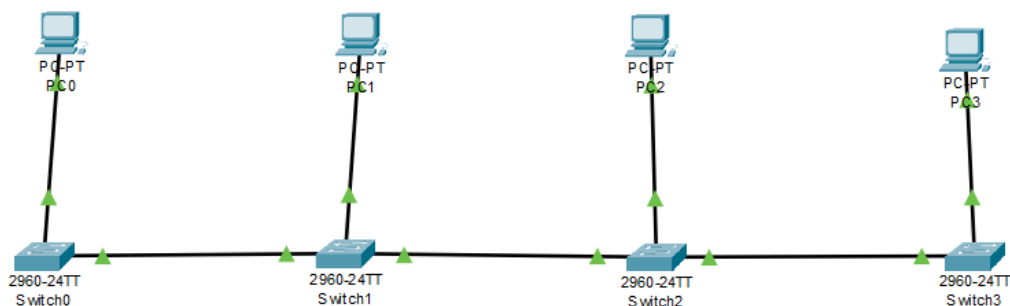
- Set up and simulate different network topologies.
- Identify the strengths and weaknesses of each topology.
- Compare network efficiency, reliability, and failure handling.

Description:

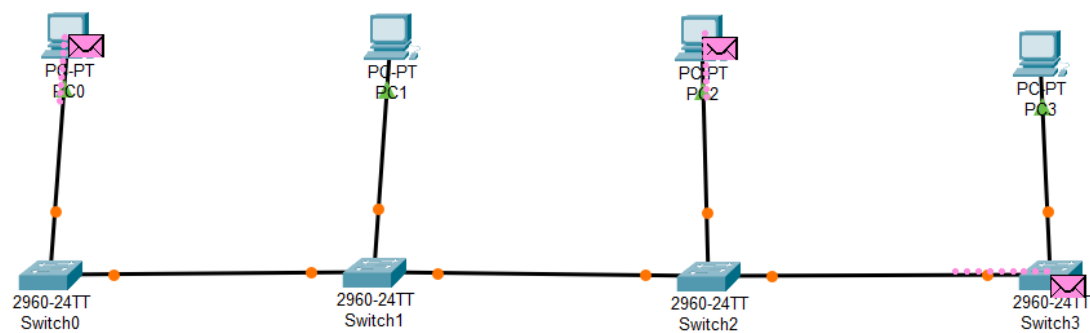
Simulate and compare Bus, Star, Mesh, and Ring topologies using a network simulation tool, analyze their advantages and disadvantages, and document the results.

Bus Topology:

Setup



Simulation



IP Addressing Test

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

Control-C
^C
C:\>
C:\>ping 198.168.1.1

Pinging 198.168.1.1 with 32 bytes of data:

Reply from 198.168.1.1: bytes=32 time=24ms TTL=128
Reply from 198.168.1.1: bytes=32 time=6ms TTL=128
Reply from 198.168.1.1: bytes=32 time=7ms TTL=128
Reply from 198.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 198.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 24ms, Average = 9ms
C:\>|
```

☐ Top

Simulation Panel

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	16.969	Switch3	PC3	DTP
	16.969	Switch1	Switch0	DTP
	16.969	Switch2	Switch3	DTP
	16.969	Switch0	Switch1	STP
	16.969	Switch1	Switch2	DTP
	16.969	Switch2	PC2	STP
	16.969	--	Switch0	DTP
	16.970	Switch2	Switch1	DTP
	16.970	Switch3	Switch2	STP
	16.970	Switch1	PC1	STP
	16.970	Switch1	Switch0	STP
	16.970	Switch2	Switch3	STP
	16.970	Switch0	Switch1	DTP
	16.971	Switch2	Switch1	STP
	16.971	Switch0	PC0	STP
	16.971	Switch3	PC3	STP
	17.968	--	Switch2	STP
	17.969	Switch2	Switch3	STP
	17.969	--	Switch1	STP
Visible	17.970	--	Switch1	STP
Visible	17.970	Switch1	Switch0	STP
Visible	17.970	--	Switch3	STP

Reset Simulation ☒ Constant Delay Captured to: 17.970 s

Play Controls

Event List Filters - Visible Events

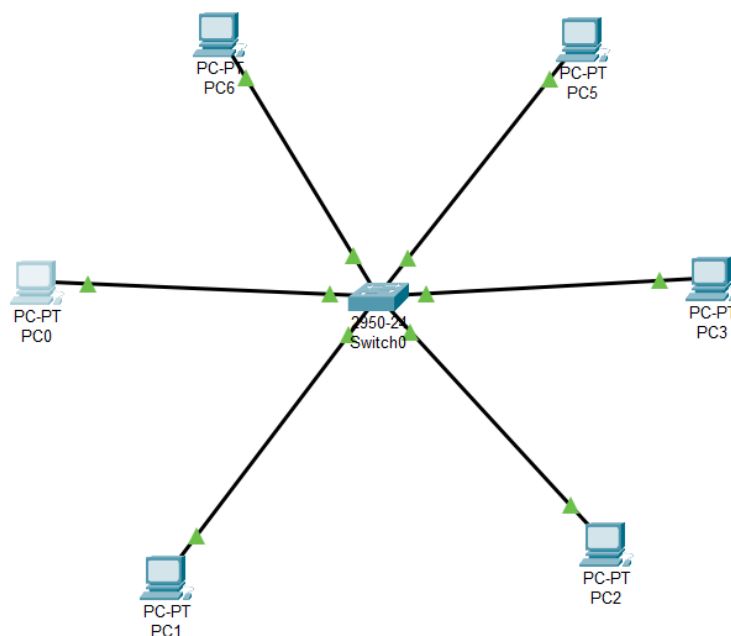
ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DNS, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters Show All/None

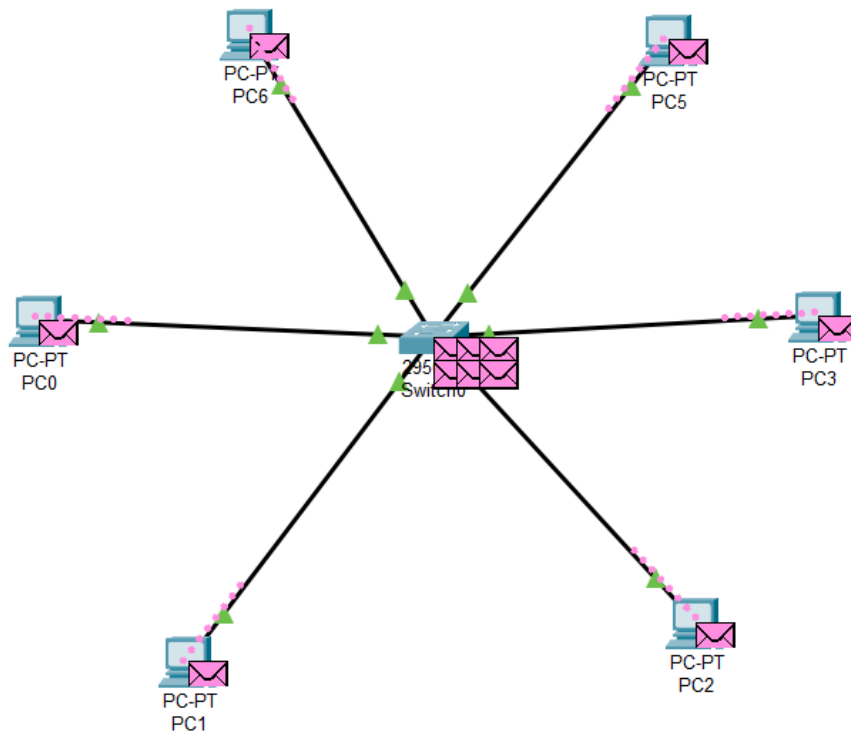
Event List Realtime Simulation

Star Topology:

Setup



Simulation



IP Addressing Test

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=3ms TTL=128
Reply from 192.168.1.1: bytes=32 time=7ms TTL=128
Reply from 192.168.1.1: bytes=32 time=6ms TTL=128
Reply from 192.168.1.1: bytes=32 time=8ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 8ms, Average = 6ms

C:\>
```

Simulation Panel

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.997	Switch0	PC1	STP
	0.997	Switch0	PC0	STP
	2.993	--	Switch0	STP
	2.994	--	Switch0	STP
	2.994	--	Switch0	STP
	2.994	--	Switch0	STP
	2.994	--	Switch0	STP
	2.994	--	Switch0	STP
	2.994	Switch0	PC2	STP
	2.994	Switch0	PC3	STP
	2.994	Switch0	PC5	STP
	2.994	Switch0	PC6	STP
	2.994	Switch0	PC1	STP
	2.994	Switch0	PC0	STP
	2.994	--	Switch0	STP
	2.995	Switch0	PC2	STP
	2.995	Switch0	PC3	STP
	2.995	Switch0	PC5	STP
	2.995	Switch0	PC6	STP
	2.995	Switch0	PC1	STP
	2.995	Switch0	PC0	STP

Reset Simulation

☒ Constant Delay

Captured to: 2.995 s

Play Controls

Event List Filters - Visible Events

ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters

Show All/None

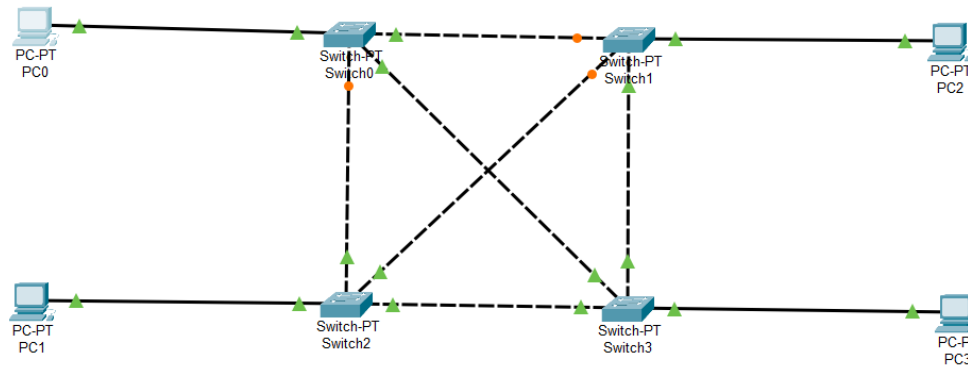
Event List

Realtime

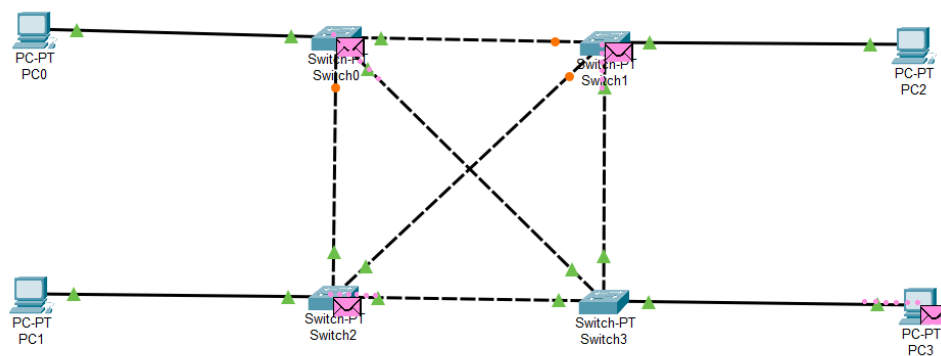
Simulation

Mess Topology:

Setup



Simulation



IP Addressing Test

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=5ms TTL=128
Reply from 192.168.1.1: bytes=32 time=8ms TTL=128
Reply from 192.168.1.1: bytes=32 time=7ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 5ms

C:\>
```

Simulation Panel

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	2.305	--	Switch0	CDP
	2.305	--	Switch0	CDP
	2.306	Switch0	PC0	CDP
	2.306	Switch0	Switch1	CDP
	2.306	Switch0	Switch2	CDP
	2.306	Switch0	Switch3	CDP
	2.995	--	Switch3	STP
	2.996	Switch3	PC3	STP
	2.996	Switch3	Switch1	STP
	2.996	Switch3	Switch2	STP
	2.996	Switch3	Switch0	STP
	2.997	Switch1	PC2	STP
	2.997	Switch1	Switch2	STP
	2.997	Switch1	Switch0	STP
	2.997	Switch2	Switch0	STP
	2.997	Switch2	PC1	STP
	2.997	Switch2	Switch1	STP
	2.997	Switch0	PC0	STP
	2.997	Switch0	Switch1	STP
	3.997	--	Switch0	STP
Visible	3.998	Switch0	Switch1	STP

Reset Simulation

☒ Constant Delay

Captured to: 3.998 s

Play Controls

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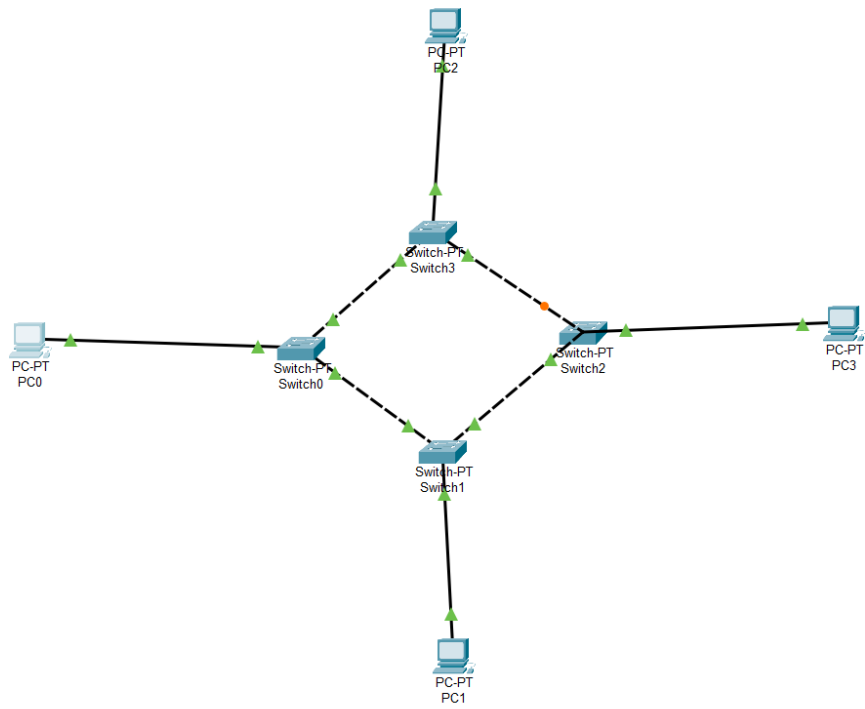
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Event List Filters - Visible Events

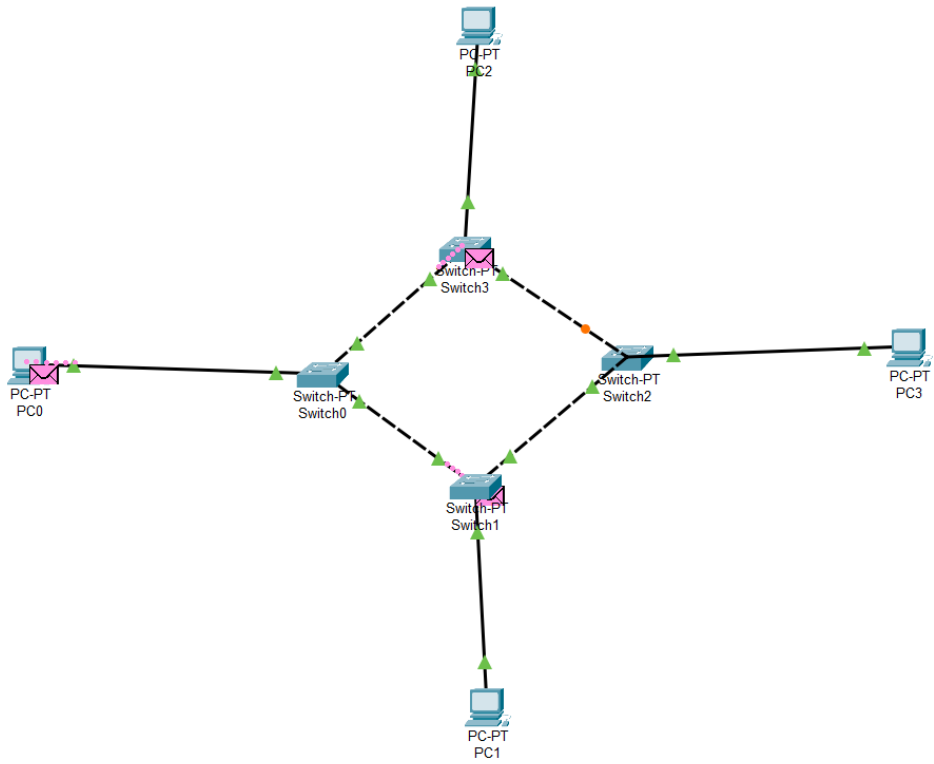
ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Ring Topology:

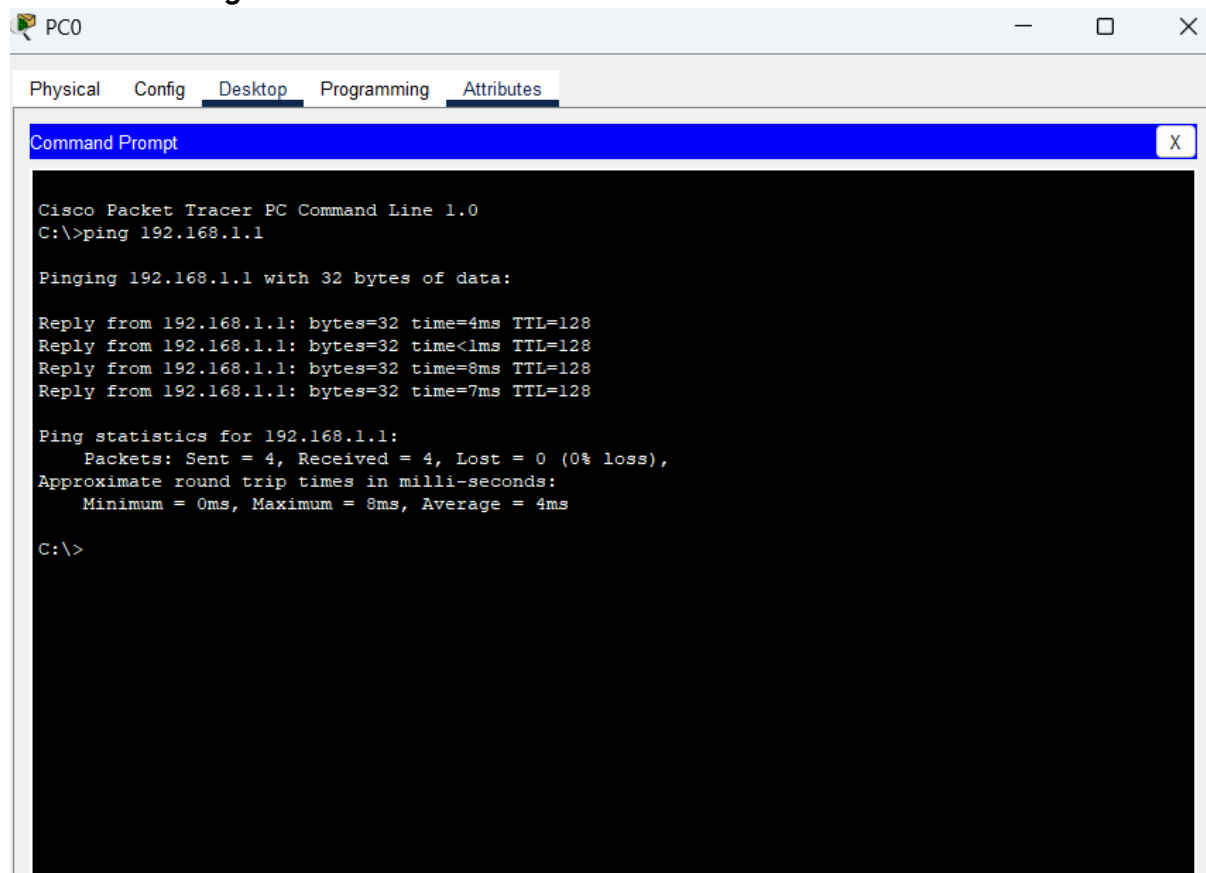
Setup



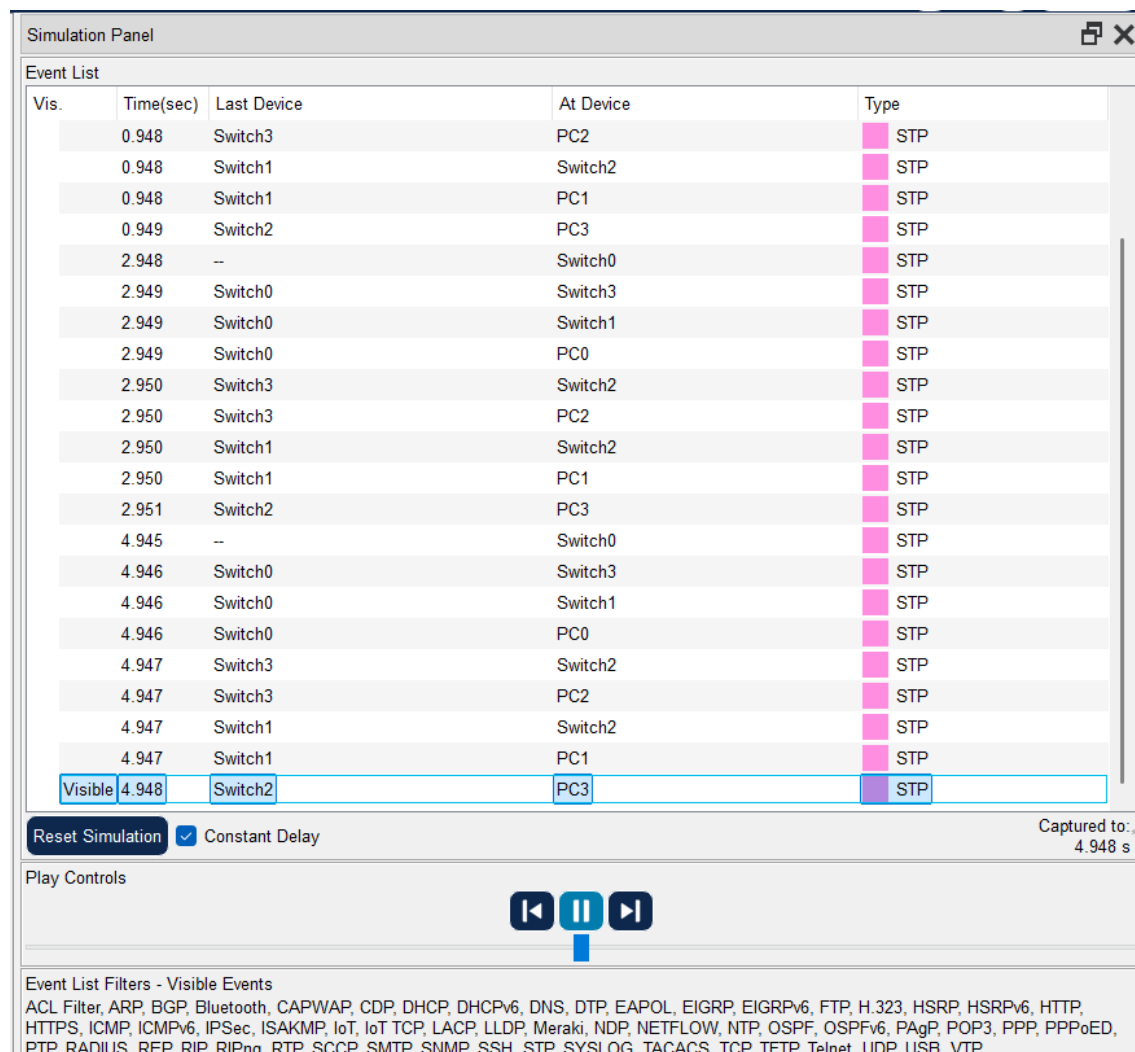
Simulation



IP Addressing Test



Simulation Panel



Comparison of Network Topologies

Topology	Latency (ms)	Data Transfer Speed	Failure Handling	Efficiency
Bus	Moderate	Low	Poor (<i>single point failure</i>)	Low
Star	Low	High	Medium (<i>central switch failure</i>)	High
Mesh	Very Low	Very High	Excellent (<i>self-healing</i>)	Very High
Ring	Moderate	Medium	Poor (<i>loop failure</i>)	Medium